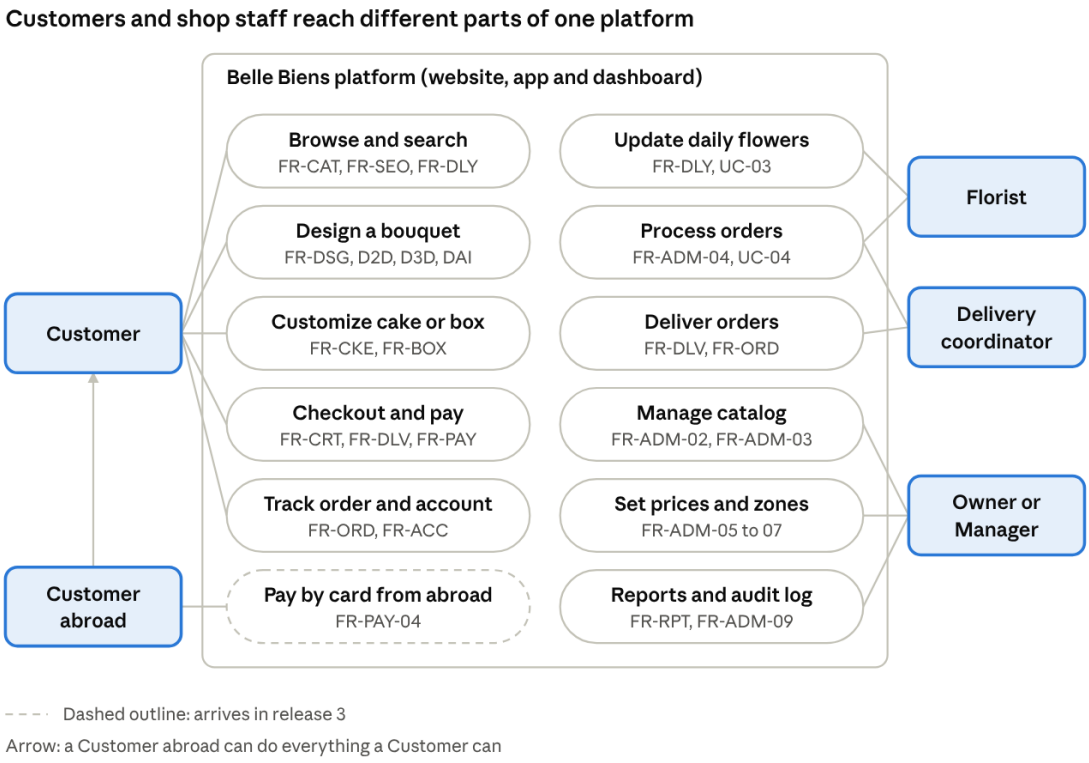
Belle Biens — Design Diagrams

Nov 10, 2026 · @حمزة

These diagrams turn the SRS (BB-SRS-001) into the structure the code will follow, and each one cites the requirement IDs it comes from.

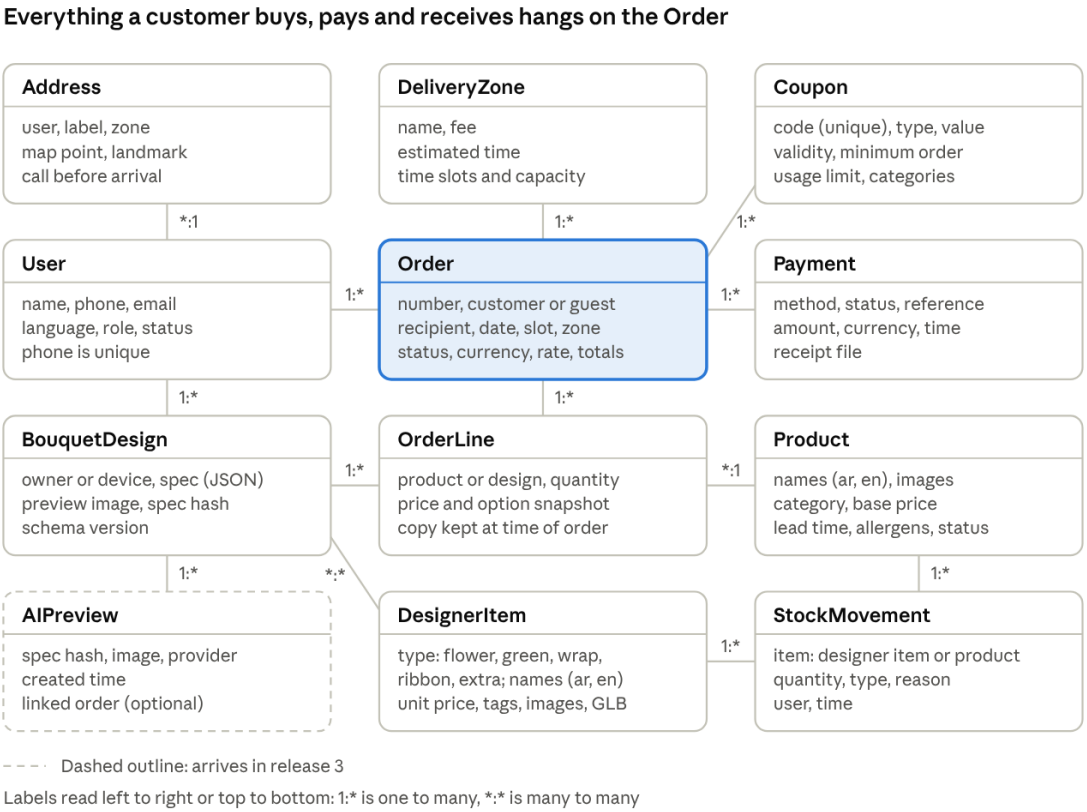
1. Use case diagram



use cases · 5 actors, 12 use case groups

Customers use the left half and staff use the right half; every use case names the SRS requirement groups behind it. الخطوط المتقطعة تعني ميزة تصل في المرحلة 3.

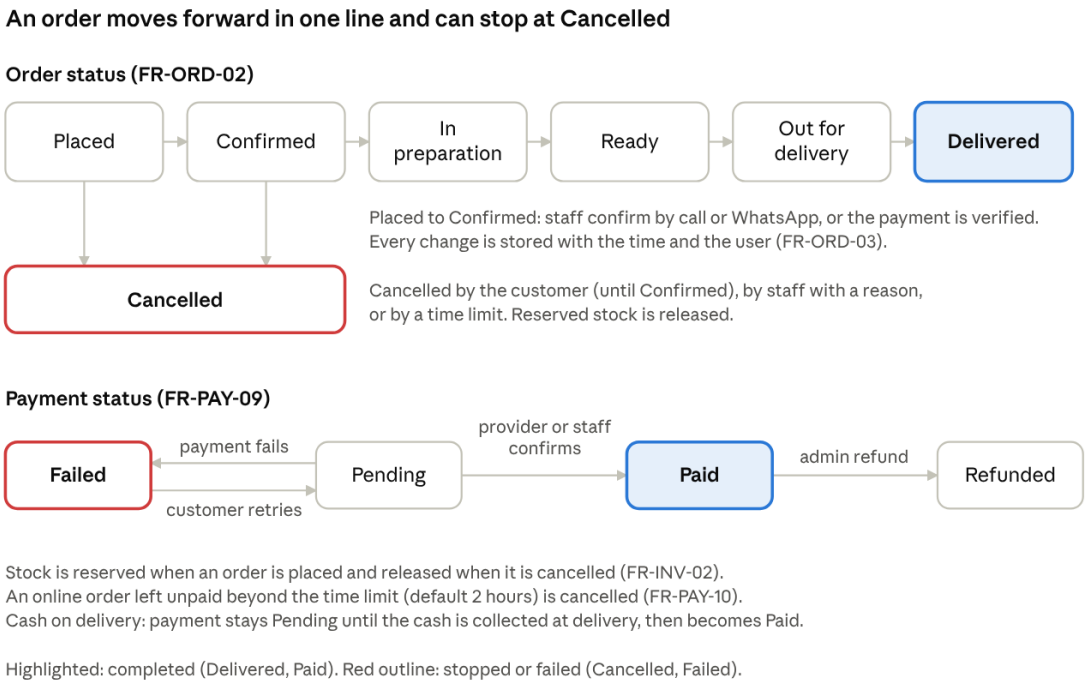
2. Data model (ERD)



data model · 12 core entities, Order at the center

The diagram shows the 12 core entities from SRS section 5.1 (DR-01 to DR-21); stock levels, audit log, settings, reviews, reminders and notification logs are left out to keep it readable. The OrderLine keeps a copy of prices and options (FR-ORD-06), and a StockMovement points to either a designer item or a product.

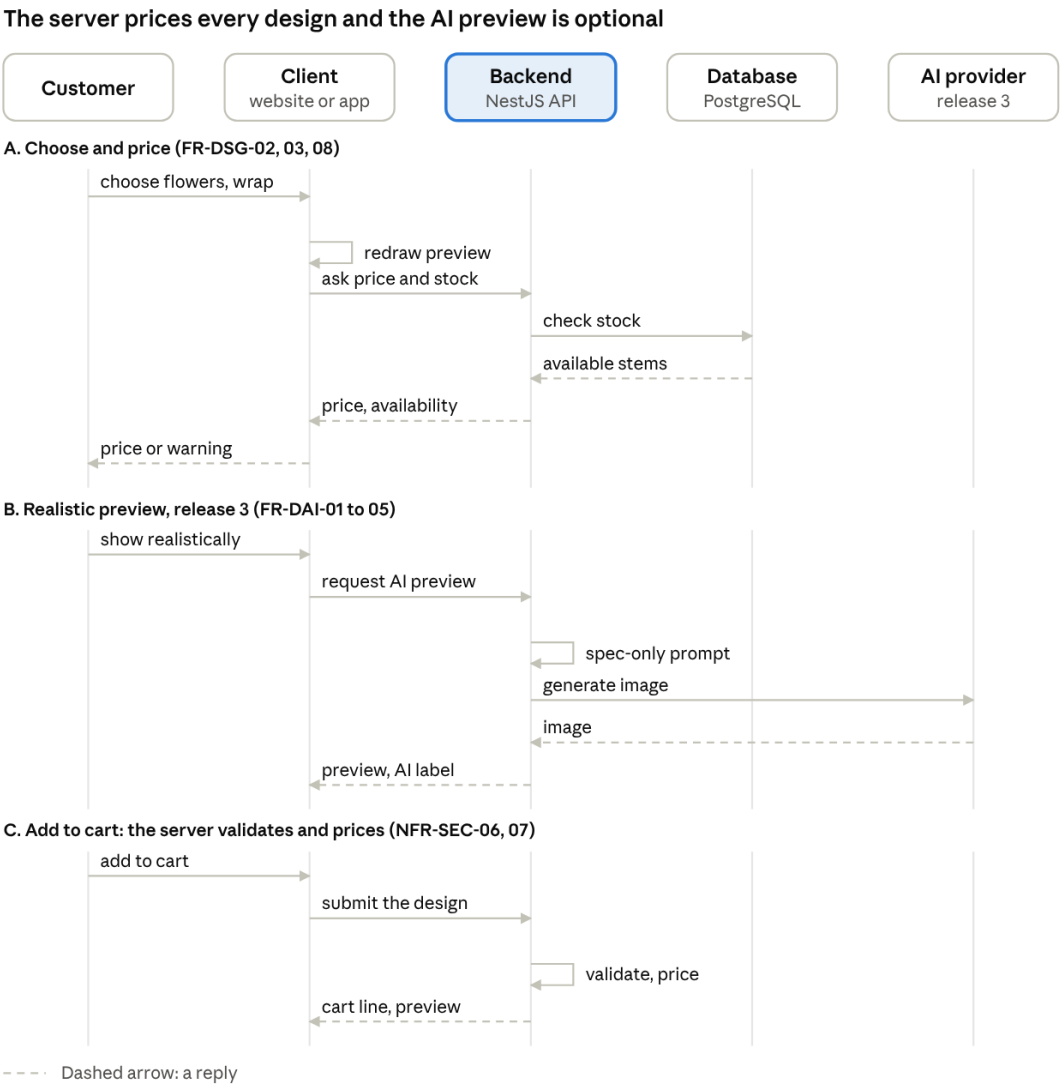
3. Order and payment lifecycle



state diagram · order status and payment status

The order and its payment move through separate states, so an order can be Confirmed while its payment is still Pending (cash on delivery). The payment values (Pending, Paid, Failed, Refunded) and the rule that cash on delivery becomes Paid at handover are my proposal; the SRS only requires a payment history and a refund record, so confirm them with the shop.

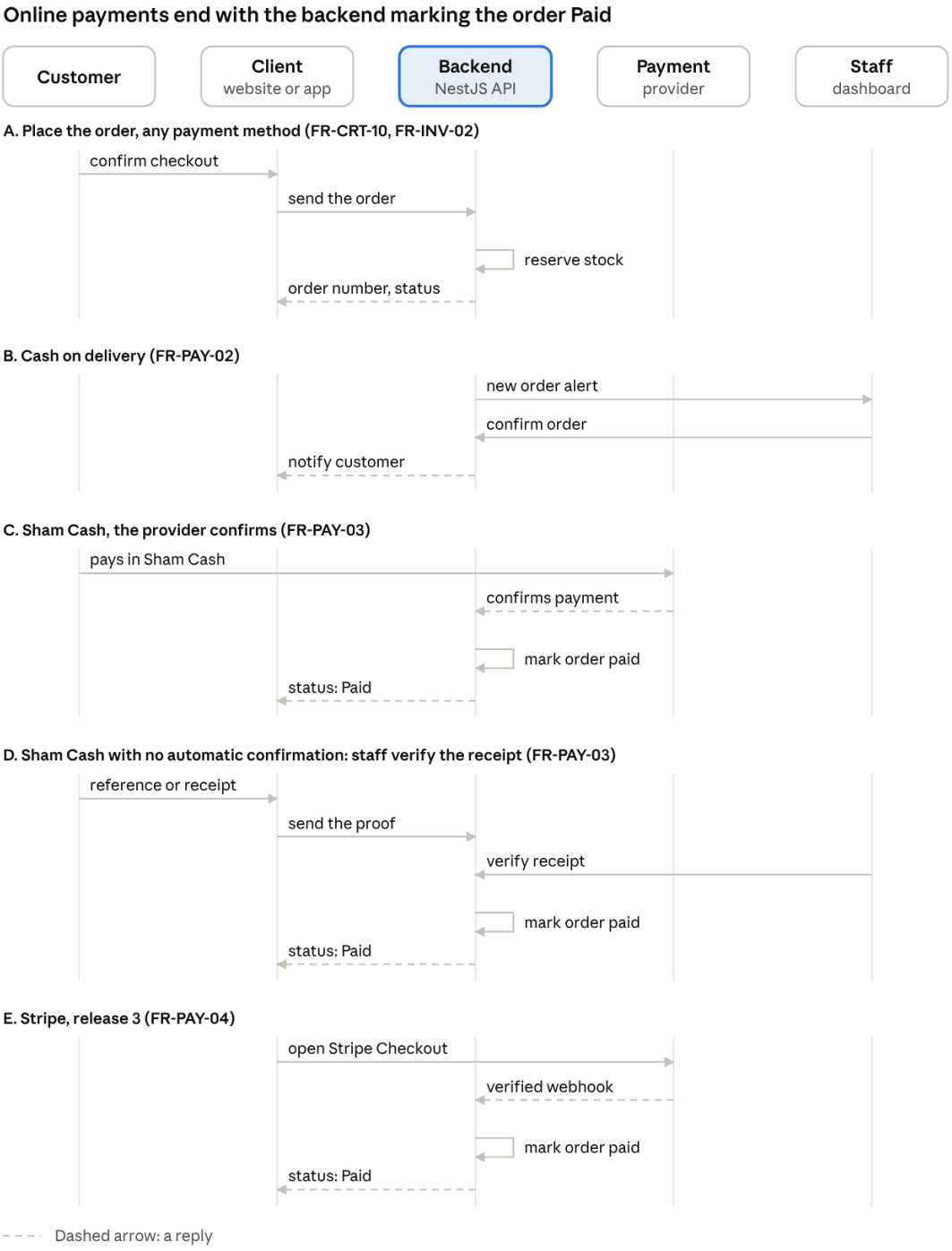
4. Bouquet designer flow



sequence · designer, price check, AI preview, add to cart

The client redraws the preview on its own, so the customer never waits for the server to see a change; the server is asked only for the price and the stock. Part B can fail or be skipped without blocking part C (FR-DAI-04, FR-DAI-08).

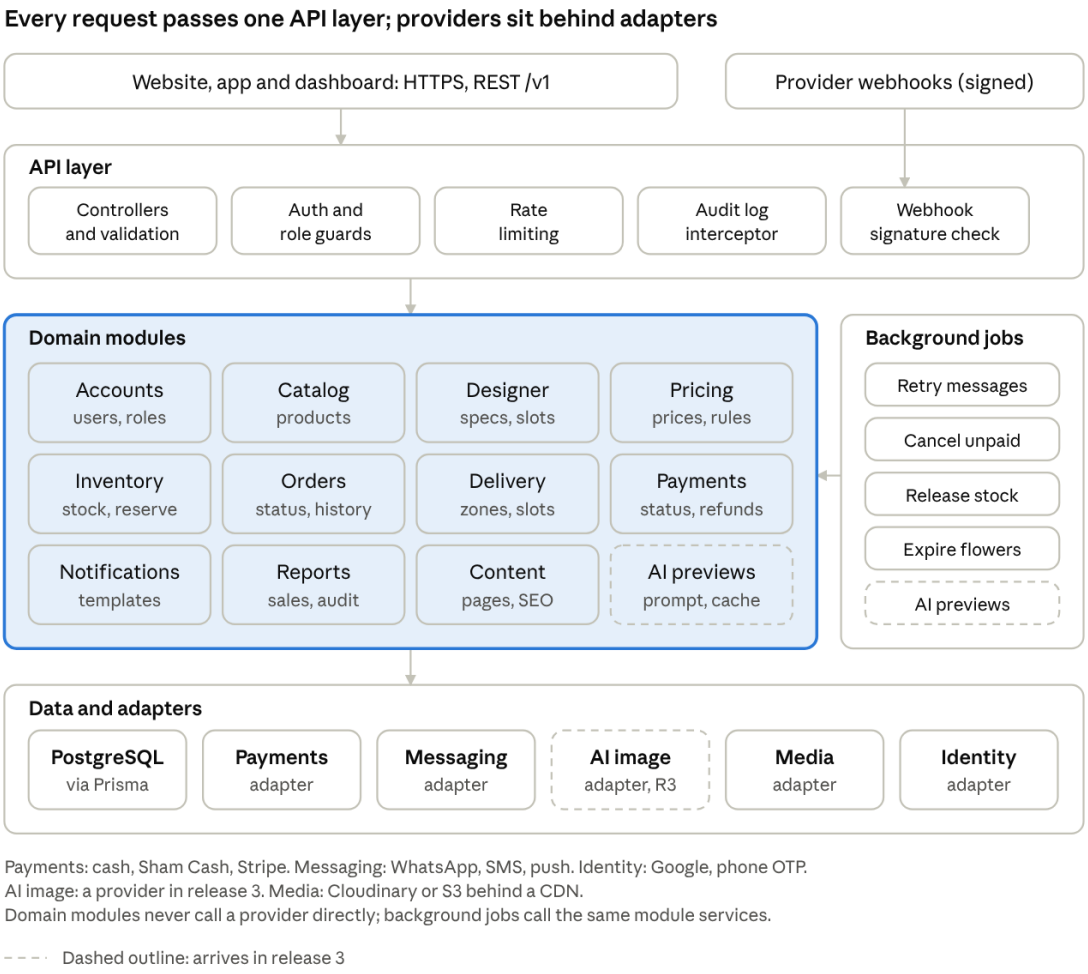
5. Checkout and payment flow



sequence · checkout with cash, Sham Cash and Stripe

Placing the order is the same for every method (A), and only the way it becomes confirmed or paid differs. For Stripe the order is marked paid only after a verified webhook, never from the browser (FR-PAY-04, EI-CM-03).

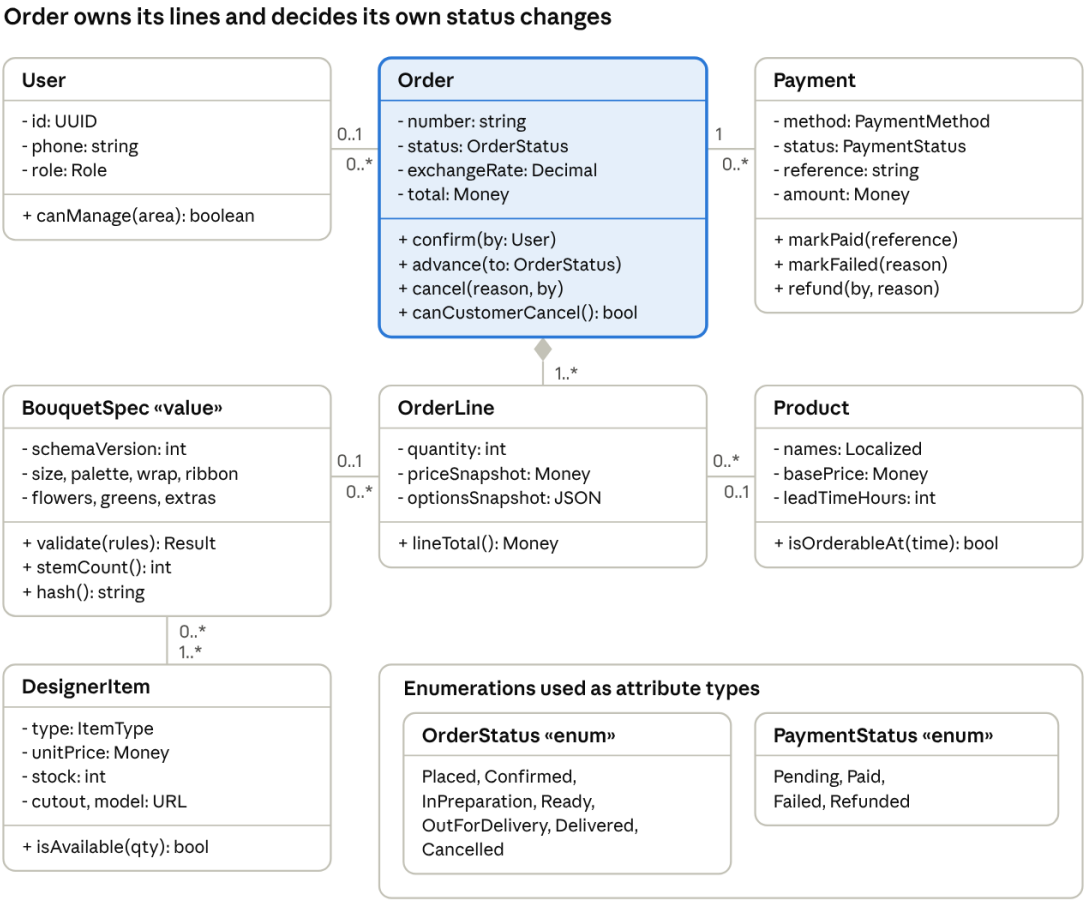
6. Backend architecture



backend architecture · layers, modules, jobs and adapters

The backend is one NestJS service split into modules (DC-01, DC-04), not separate microservices, which keeps price, stock and order rules in one place at this size. Controllers carry the checks that must never be skipped (roles, rate limits, audit log, webhook signatures), and the swap-a-provider rule of NFR-FLX-02 is met because only the adapters know a provider's API.

7. Class diagram: domain model

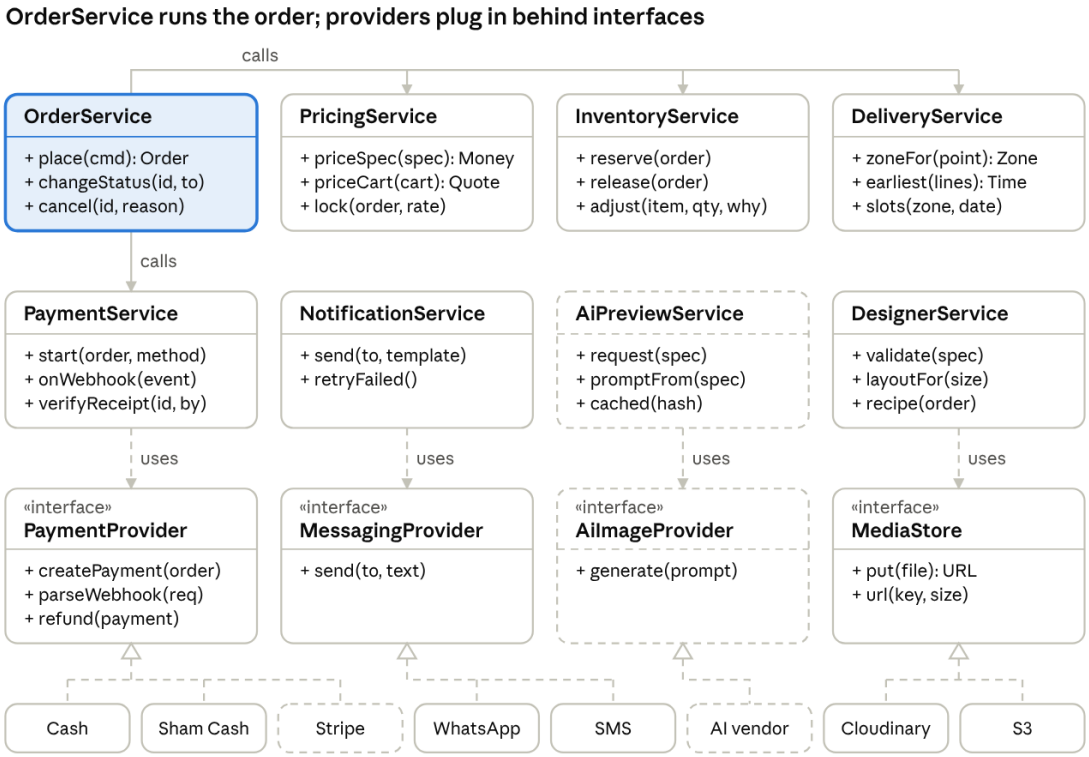


Filled diamond: composition (an order owns its lines). A dash marks an attribute, a plus an operation.
Each number sits at the end it counts; 0..1 at User means guest orders have no user.

UML class diagram · domain model, 7 classes and 2 enumerations

The classes hold their own rules: an Order only moves to a status that FR-ORD-02 allows, and a BouquetSpec validates itself against the design rules (FR-DSG-14). Money and Localized are small value types (amount with currency; Arabic and English text) used across the model.

8. Class diagram: services and providers



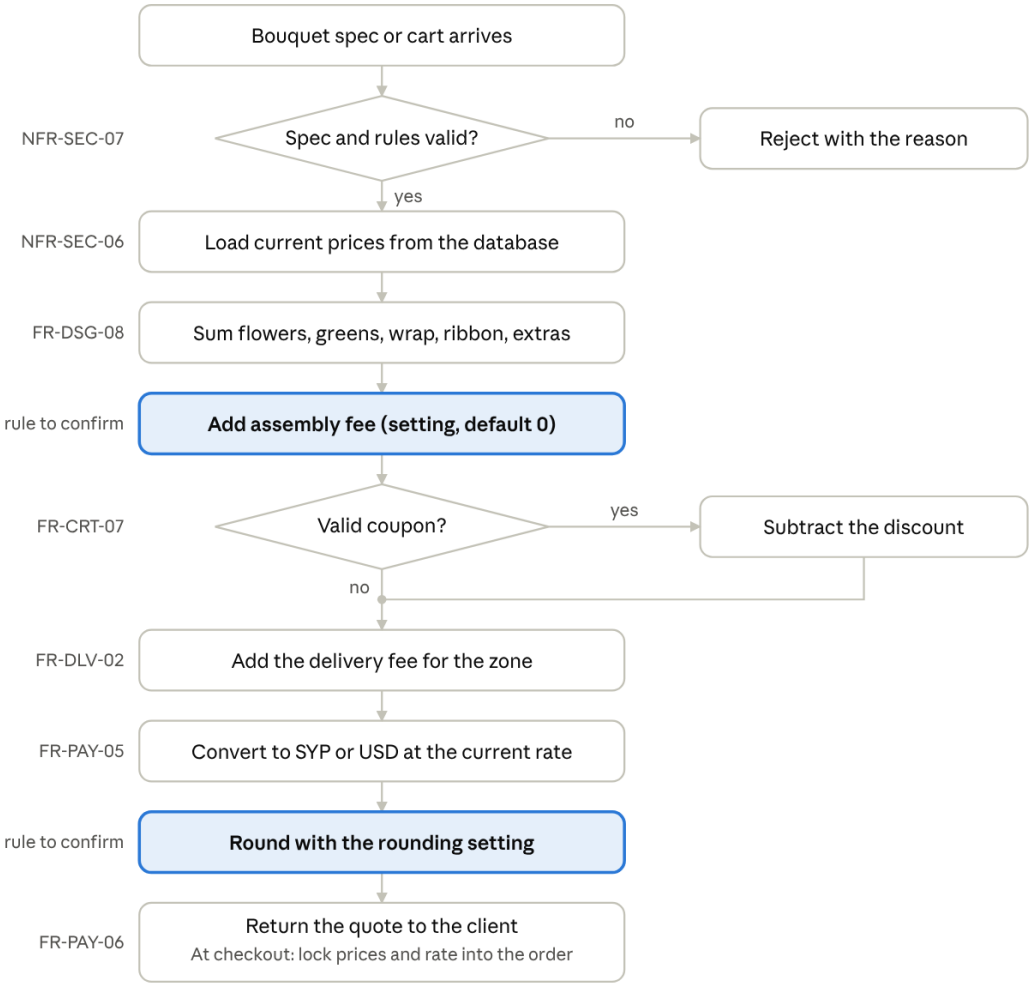
OrderService and PaymentService also call NotificationService; lines are left out to keep the picture readable.
Hollow triangle with dashed line: the class below implements the interface. Dashed outline: release 3.
The AI vendor is still open (OI-11), and push (FCM, APNs) sits behind MessagingProvider too.

UML class diagram · 8 services, 4 provider interfaces, 8 implementations

Services depend on interfaces, never on a vendor, so adding or dropping Stripe or Sham Cash means writing or removing one class (FR-PAY-07, NFR-FLX-02). Each service is the public face of one domain module in diagram 6.

9. Price calculation pipeline

The server builds every price from its own data, step by step

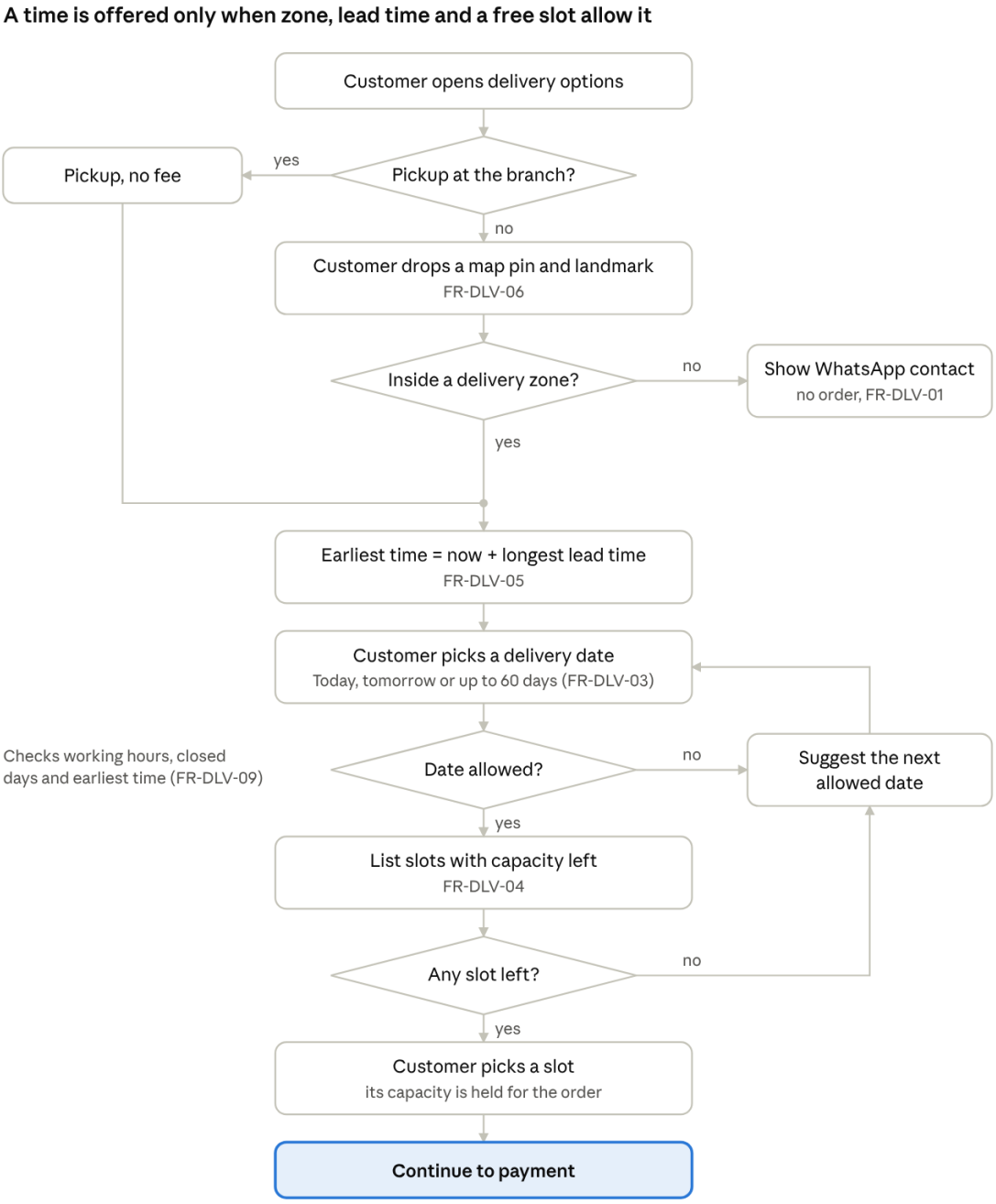


Highlighted: the value comes from an admin setting whose rule the shop still has to confirm.
Prices sent by the website or app are never used; the same steps run for a bouquet, a cake, a box and a whole cart.

flowchart · price calculation, 9 steps and 2 decisions

This is the pricing module that lets the MVP start before the shop decides its rules: the two highlighted steps read admin settings (assembly fee 0, no rounding by default), so the real rules go in later without code changes. Step order is my proposal; for example, whether the coupon applies before or after the delivery fee is a business choice to confirm.

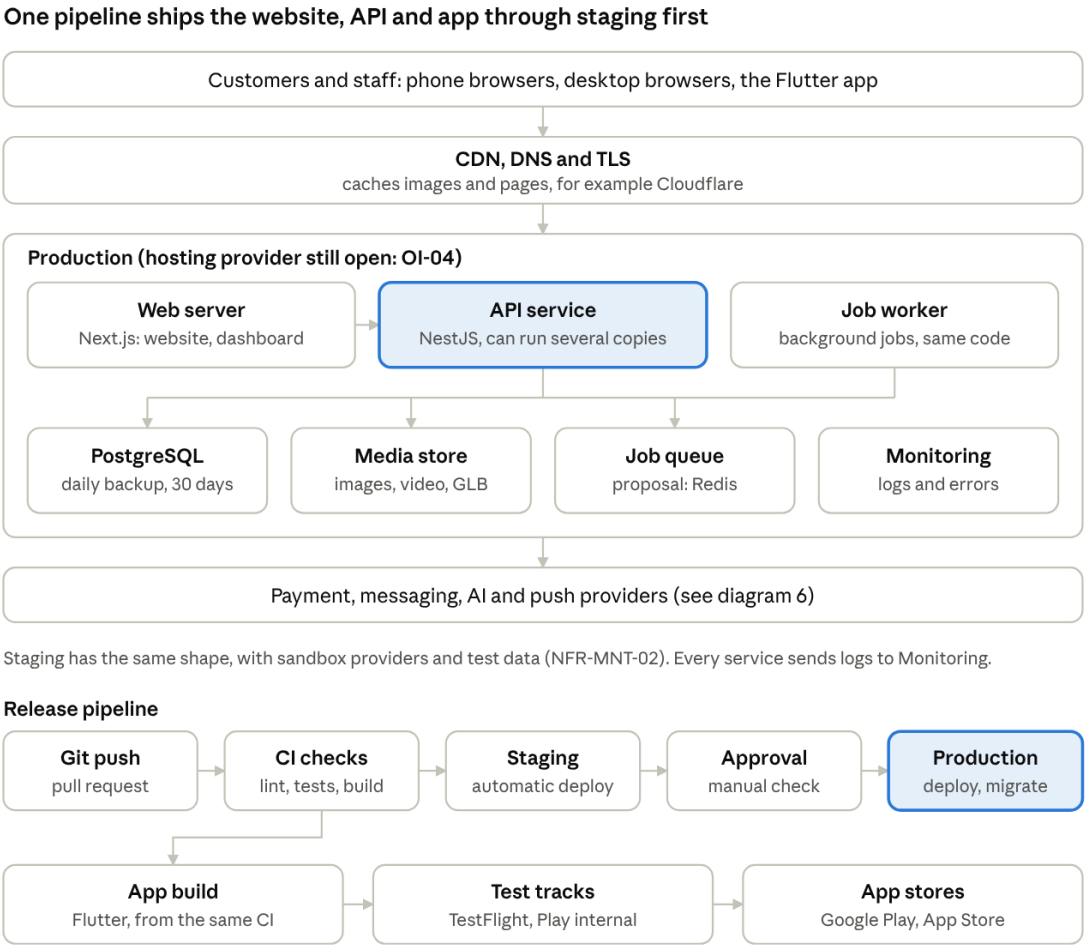
10. Delivery date and slot selection



decision flow · delivery date and time slot

The checkout never shows a time the shop cannot meet, which is the main source of complaints for same-day flowers. Holding a slot's capacity until the order is placed or expires is my proposal (FR-DLV-04 says only that a full slot is not offered).

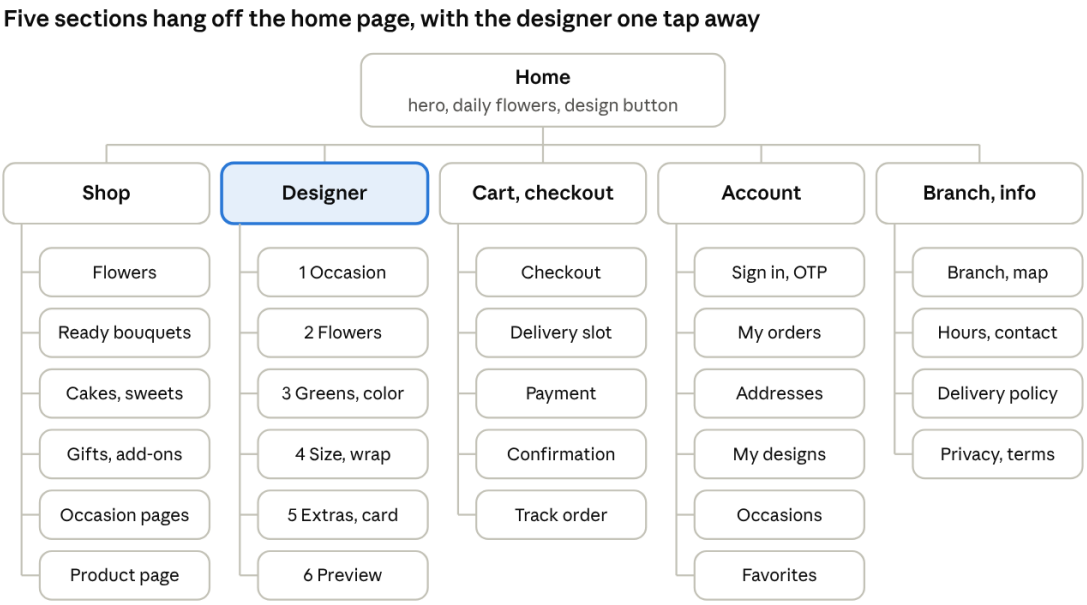
11. Deployment and release pipeline



deployment · production tiers and release pipeline

The job worker runs the same code as the API but only processes background jobs, so a slow AI request or a retry storm never slows the checkout. The job queue (Redis) and Cloudflare are proposals, not SRS decisions, and the hosting provider is still open item OI-04.

12. Customer screen map

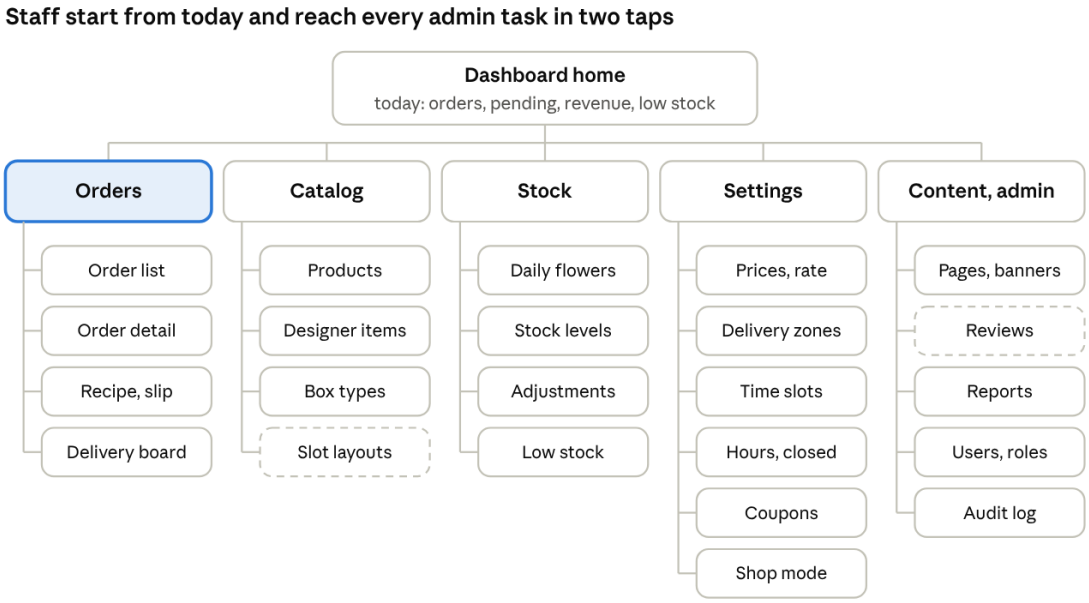


The website and the Flutter app share this map. Every screen header has search, language switch, cart and WhatsApp.
Step 6 switches the preview between 2D, 3D and AI; 3D and AI arrive in release 3.
Guests reach Track order with the order number and phone (FR-ORD-04); My designs, Occasions and Favorites need an account.

screen map · customer website and app, 5 sections, 27 screens

This is the list the wireframes start from: each box becomes one screen or one step. الترتيب من اليسار إلى اليمين هنا للقراءة فقط؛ في الواجهة العربية ينعكس الاتجاه (FR-LOC-01).

13. Dashboard map and roles



screen map · admin dashboard, 5 sections, 23 screens

The roles come from FR-ADM-01; who may do what is my proposal for the shop to confirm. The server enforces this table on every API call (NFR-SEC-01), so hiding a screen is a convenience, not the protection.

Area	Owner	Manager	Florist	Delivery coordinator
Dashboard home	Yes	Yes	Yes	Yes
Orders: view and change status	Yes	Yes	Yes	Delivery statuses only
Orders: cancel, refund	Yes	Yes	No	No
Print recipe and order slip	Yes	Yes	Yes	Yes
Daily flowers and stock adjustments	Yes	Yes	Yes	No
Products, designer items, box types	Yes	Yes	No	No
Prices, exchange rate, coupons	Yes	Yes	No	No
Delivery zones, slots, hours, shop mode	Yes	Yes	No	View only

Area	Owner	Manager	Florist	Delivery coordinator
Pages, banners, reviews	Yes	Yes	No	No
Reports	Yes	Yes	No	No
Users and roles	Yes	No	No	No
Audit log	Yes	View only	No	No